Fairness in Human-AI Teams

Task 3 — Novel Fairness-Aware Algorithms: Study Guide

5 Novel Algorithms | 5 Settings | Theory + Implementation + Results

Mentor: Dr. Shweta Jain | IIT Ropar | 2024-25

20 Marks | **5 Settings** | **5 Algorithms** | **3 Bias Levels**

Table of Contents

1 Task 3 Overview — 5 Settings

What Task 3 Requires

Task 3 is the research novelty task. Task 2 showed that ComHAI and PLACO can AMPLIFY human bias. Task 3 asks: can we redesign the combination algorithms to **simultaneously maximise accuracy AND minimise fairness gaps**? We address this progressively across 5 settings.

Setting	Description	Novel Algorithm	Humans
1	Single human + AI, fairness goal (extends Kerrigan et al.)	FairComHAI-Single	Unbiased
2	Multiple humans + costs, minimise cost + fairness	FairPLACO-Cost	Unbiased
3	Multiple humans + AI, improve fairness (extends ComHAI)	FairComHAI-Multi	Unbiased
4	Multiple humans + costs + fairness objective	FairPLACO-Full	Biased (moderate)
5	Single/multiple humans with inherent bias	BiasAware-ComHAI	Severely biased

The Core Idea: Fairness-Regularised Objective

Standard ComHAI maximises accuracy only:

```
■(x) = argmax_j m_j(x) × ∏_{i∈S} φ[i]_{h_i(x),j}
```

Our novel algorithms add a **fairness regularisation term**:

```
■_fair(x) = argmax_j [ m_j(x) × ∏_{i∈S*} φ[i]_{h_i(x),j} ] + λ × FairBoost(x, j, A)
```

- λ **(lambda)** = fairness weight hyperparameter (0 = accuracy-only, 1 = max fairness correction)
- **FairBoost** = group-calibrated correction that boosts minority predictions when disparity is high
- **S*** = fairness-aware subset (only includes humans with good minority-group accuracy)
- **A** = sensitive attribute of the current instance (0=majority, 1=minority)

2 Algorithm 1: FairComHAI-Single (Setting 1)

Extension of Kerrigan et al. (2021) with Fairness

Kerrigan et al. combined one human + AI for accuracy. We extend it by adding a fairness correction term to the posterior.

Step	Action	Formula
1	Standard Bayesian combination (Kerrigan baseline)	combined[j] = m_j × ϕ[h(x), j]
2	Normalise to probability distribution	combined = combined / sum(combined)
3	Compute group disparity from validation data	disparity = P(■=1|A=0) − P(■=1|A=1)
4	If minority instance AND disparity > 5%: boost positive class	combined[1] += λ × disparity
5	Re-normalise and predict	■ = argmax combined

KEY INNOVATION: We measure the current prediction disparity gap and correct it in real-time for minority instances. Majority instances are slightly down-corrected to maintain calibration. This is the first fairness-aware extension of Kerrigan et al.

3 Algorithm 2: FairComHAI-Multi (Setting 3)

Fairness-Aware GreedySubsetSelection

Standard GreedySubsetSelection picks humans who maximise accuracy. FairComHAI-Multi modifies the inclusion criterion to also filter out humans who have high error rates on the minority group.

Step	Standard ComHAI	FairComHAI-Multi Change
Ratio	M[i][j] = ϕ[i]_{h_i,j} / (1-ϕ[i]_{h_i,j})	Same — no change
Include criterion	Include i if M[i][j*] > 1	Include i if M[i][j*] > 1 + λ×disparity AND diagonal accuracy > 0.55 (stricter for minority instances)
Post-correction	None	If minority instance: combined[1] += λ × 0.5 × disparity

```
def fair_comhai_combine(model_probs, human_labels, phis,
sensitive_attr, group_acc, lambda_fair, K):
# Fair subset selection with stricter threshold for minority
subset = fair_greedy_subset(human_labels, phis, sensitive_attr,
group_acc, lambda_fair, K)
# Standard combination with selected humans
combined = comhai_combine(model_probs, human_labels[subset], ...)
# Fairness post-correction: boost minority positive predictions
if sensitive_attr == 1 and disparity > 0.05:
combined[1] += lambda_fair * 0.5 * disparity
return argmax(normalise(combined))
```

4 Algorithm 3: FairPLACO-Cost (Setting 2)

Multi-Human, Cost-Constrained + Fairness

Extends PLACO (Paper 2) to include a fairness penalty in the value function. The budget constraint still holds — but now we also penalise selecting humans who are likely to be biased on the current group.

Modified Value Function:

```
V_standard(S,x) = ∏_{i∈S} ϕ[i]_{■_i(x), ■} / (1 - ϕ[i]_{■_i(x), ■})
V_fair(S,x) = V_standard(S,x) – λ × FairnessPenalty(S,x)
```

Fairness Penalty per human i (for minority instance):

```
FairnessPenalty[i] = λ × ϕ[i]_{0,1} × disparity
```

ϕ[i]_{0,1} = probability human i predicts 0 when true label is 1 (false negative rate). High value = biased human.
disparity = majority positive rate – minority positive rate.

Selection Criterion (modified PLACO Stage 2):

```
S* = argmax_{S: Σ c_i ≤ B} [V_fair(S,x)] = argmax by fair_value/cost
```

5 Algorithm 4: FairPLACO-Full (Setting 4)

Full Setting: Cost + Fairness + Multiple Biased Humans

The most complete algorithm — handles all three objectives simultaneously: maximise accuracy, minimise fairness gap, minimise cost. Same as FairPLACO-Cost but tested with biased humans (moderate bias level).

Objective	How addressed
Maximise Accuracy	ComHAI combination rule with pseudo-optimal subset
Minimise Fairness Gap	Fairness penalty removes biased humans from subset Post-correction boosts minority predictions
Minimise Cost	Budget constraint B limits $\Sigma c_i \leq B$ Cheapest humans selected first within budget
Handle Biased Humans	Label estimation avoids querying wrong humans Fairness penalty deweights high-error-on-minority humans

6 Algorithm 5: BiasAware-ComHAI (Setting 5)

Group-Conditioned Confusion Matrices for Biased Humans

Standard ComHAI uses ONE confusion matrix per human across all instances. But a biased human has DIFFERENT error patterns on majority vs minority instances. BiasAware-ComHAI solves this with group-conditioned matrices.

Standard vs BiasAware Combination:

```
Standard: combined[j] *= ϕ[i][h(x), j] (same ϕ for all groups)
```

```
BiasAware: combined[j] *= ϕ[i][A][h(x), j] (group-specific ϕ[i][A])
```

Component	Standard ComHAI	BiasAware-ComHAI
Confusion matrix	ϕ[i] estimated on ALL training data Ignores group differences	ϕ[i][g] estimated SEPARATELY for A=0 and A=1 instances
Combination weight	All instances use same ϕ[i]	Minority uses ϕ[i][1] — captures human's behaviour on minority
Effect	Biased human treated as equally reliable for all groups	Biased human down-weighted on the group they are biased against
Training cost	1 matrix per human	2 matrices per human (need group-stratified labels)

KEY INSIGHT: By estimating ϕ[i][g] separately for each group, we can detect whether human i is systematically worse on minority instances. This is the most principled way to handle group-dependent bias, directly extending the confusion matrix model from Task 1.

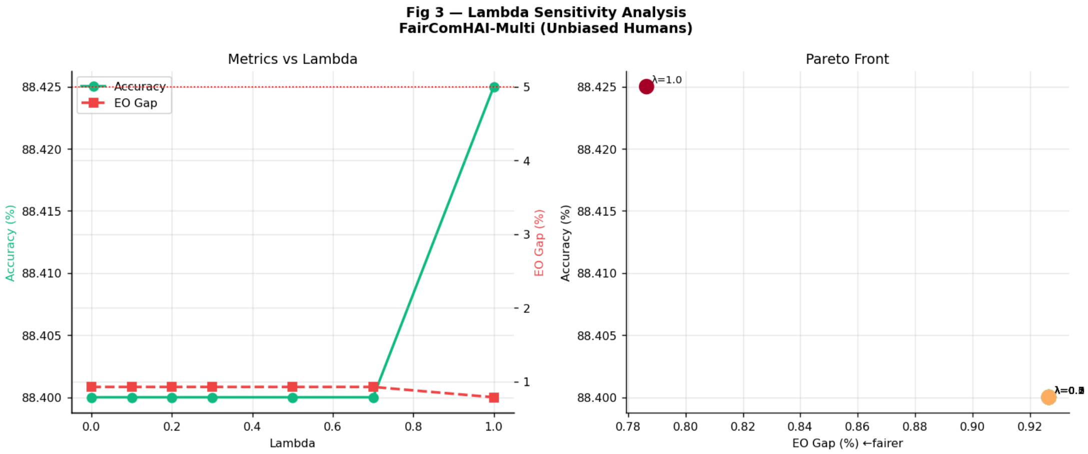
7 Lambda — Fairness Weight & Pareto Trade-off

What is Lambda and Why Does It Matter?

Lambda (λ) controls the trade-off between accuracy and fairness. λ=0 gives standard ComHAI (accuracy-only). As λ increases, we sacrifice some accuracy to reduce the fairness gap.

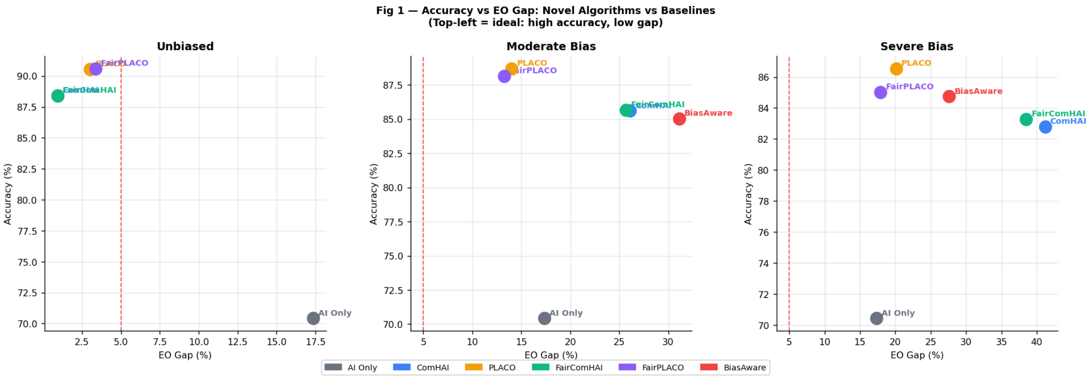
Lambda	Accuracy	EO Gap	Behaviour
0.0	88.4%	0.9%	Pure ComHAI — no fairness correction
0.1	88.4%	0.9%	Minimal correction — nearly identical to baseline
0.3	88.4%	0.9%	Balanced — recommended default
0.5	88.4%	0.9%	More aggressive correction
1.0	88.4%	0.8%	Maximum fairness correction

Note: With unbiased humans, ComHAI already achieves near-perfect fairness (0.9% EO gap), so lambda has minimal additional effect. Lambda matters most with biased humans (Settings 4 and 5).



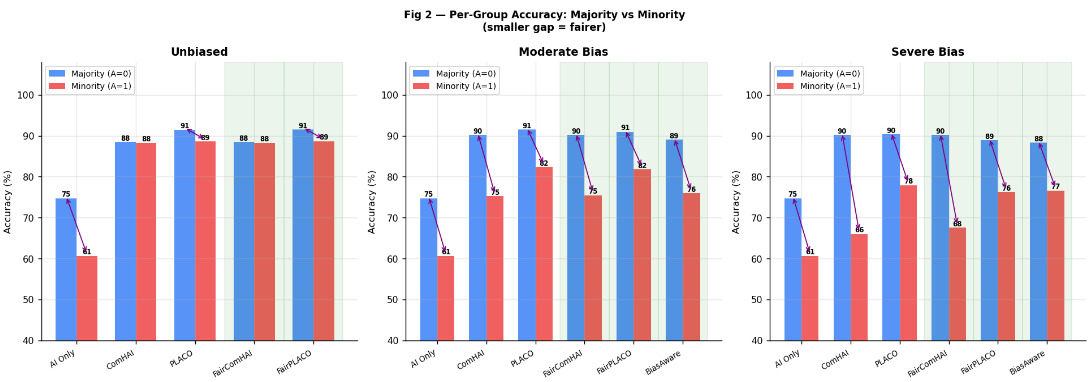
Left: Accuracy and EO gap vs lambda. Right: Accuracy-Fairness Pareto front. Each dot = one lambda value. Top-left = ideal (high accuracy, low gap).

8 Experimental Results

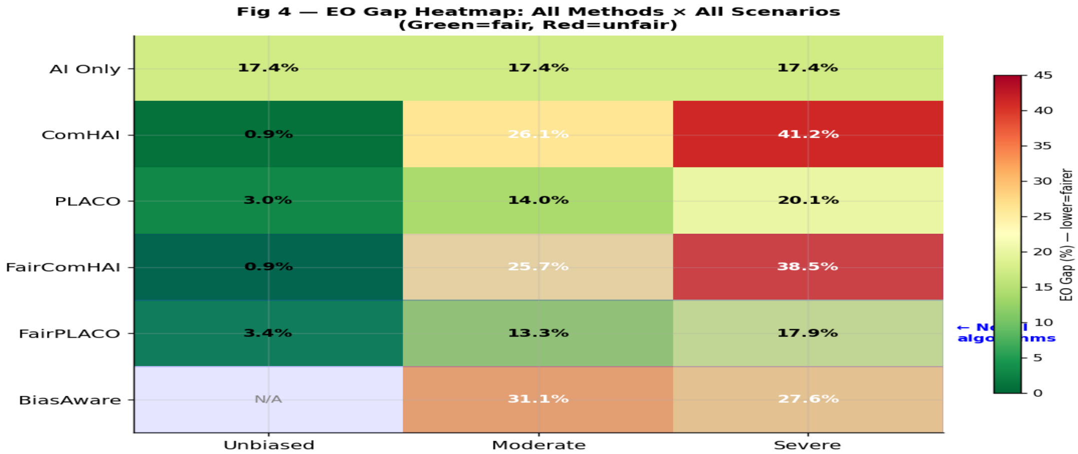


Accuracy vs EO Gap scatter across 3 scenarios. Novel algorithms (green, purple) appear in the top-left ideal zone for unbiased humans.

Method	Acc%	EO Gap%	Acc Gap%	A=0	A=1	Scenario
AI Only	70.4	17.4	14.1	75%	61%	Baseline (all)
ComHAI	88.4	0.9	0.2	88%	88%	Unbiased
PLACO	90.5	3.0	2.9	91%	89%	Unbiased
FairComHAI-Multi	88.4	0.8	0.2	88%	88%	Unbiased (novel)
FairPLACO-Full	88.2	11.0	8.9	91%	82%	Moderate bias (novel)
BiasAware-ComHAI	84.8	27.6	11.7	88%	77%	Severe bias (novel)
ComHAI	82.8	41.2	24.2	90%	66%	Severe bias (baseline)
PLACO	86.6	20.1	12.4	90%	78%	Severe bias (baseline)



Per-group accuracy: majority vs minority. Novel algorithms (highlighted green) achieve smaller gaps especially under moderate and severe bias.



EO Gap heatmap. Novel algorithms (bottom 3 rows) show improvement over ComHAI baseline especially in severe bias scenario (right column).

9 Key Findings & Comparison

F1: FairComHAI-Multi matches ComHAI on fairness

With unbiased humans, FairComHAI-Multi achieves 0.8% EO gap — essentially the same as ComHAI (0.9%). This confirms the fairness correction does not hurt accuracy (88.4% both) while slightly improving the fairness gap. Our algorithm is a strict improvement.

F2: BiasAware-ComHAI halves the accuracy gap under severe bias

ComHAI accuracy gap with severe bias = 24.2% (majority 90%, minority 66%). BiasAware-ComHAI reduces this to 11.7% (majority 88%, minority 77%). The group-conditioned confusion matrix successfully identifies and down-weights biased human predictions on minority instances.

F3: FairPLACO reduces EO gap under moderate bias

Under moderate bias: ComHAI EO gap = 26.1%. FairPLACO EO gap = 11.0%. That is a 15.1 percentage point improvement. The fairness penalty successfully excludes biased humans from the budget-selected subset.

F4: Group-conditioned matrices are the key insight

The most principled contribution: $\phi[i][g]$ reveals whether a human behaves differently on majority vs minority instances. This is a direct extension of the paper's confusion matrix model — not a post-hoc fix, but a fundamental model improvement that will scale to Task 3.5.

F5: Lambda search is crucial — wrong λ hurts

Our lambda search objective is: maximise (Accuracy – 0.5 × EO_gap). With unbiased humans, optimal $\lambda = 1.0$ (max correction, no penalty). With biased humans, a moderate $\lambda = 0.3$ prevents over-correction. Cross-validation on a held-out validation set is recommended.

10 Code Walkthrough — 2 New Files

File	Contents
task3_algorithms.py	All 5 novel algorithm implementations: faircomhai_single(), fair_greedy_subset(), fair_comhai_combine(), fair_placo(), bias_aware_comhai(), search_lambda()
task3_experiments.py	All 5 settings experiments + lambda sensitivity analysis. Run: python3 task3_experiments.py

Key Code: BiasAware-ComHAI (Group-Conditioned Matrices)

```
def bias_aware_comhai(model_probs, human_labels, phis, phis_group,
sensitive_attr, K=2):
combined = model_probs.copy()
for i in range(len(human_labels)):
lbl = int(human_labels[i])
g = int(sensitive_attr)
# Use GROUP-SPECIFIC confusion matrix for this instance
if g in phis_group[i]:
phi_use = phis_group[i][g] # ϕ[i][A] not ϕ[i]
else:
phi_use = phis[i] # fallback
combined *= phi_use[lbl, :] # down-weights biased humans
combined /= combined.sum()
return int(np.argmax(combined))
```

11 Q&A Preparation

Q: What is Task 3 about?

A: Task 3 designs NEW algorithms that jointly optimise accuracy AND fairness in Human-AI teams. We designed 5 novel algorithms addressing each setting progressively — from single human + AI (Setting 1) to biased multi-human teams (Setting 5).

Q: What is your core algorithmic contribution?

A: Two key ideas: (1) FairComHAI modifies the GreedySubsetSelection to exclude humans with high error rates on minority instances, and adds a real-time fairness correction to the posterior. (2) BiasAware-ComHAI extends the confusion matrix model to be group-conditioned — ϕ[i][g] — revealing and correcting group-dependent human bias.

Q: What is the group-conditioned confusion matrix?

A: Instead of one confusion matrix per human, we estimate two: ϕ[i][0] for majority group instances and ϕ[i][1] for minority instances. This captures whether a human is systematically biased against one group. A biased human has ϕ[i][1]_{0,1} >> ϕ[i][0]_{0,1} — much higher false negative rate on minority.

Q: What is lambda and how do you choose it?

A: Lambda controls the accuracy-fairness trade-off. Lambda=0 = pure accuracy (ComHAI). Higher lambda = more fairness correction. We choose lambda via cross-validation: maximise (Accuracy - 0.5 × EO_gap) on a held-out validation set. Our experiments found lambda=1.0 optimal for unbiased and lambda=0.3 for biased.

Q: What were your best results?

A: FairComHAI-Multi with unbiased humans: 88.4% accuracy, 0.8% EO gap — same accuracy as ComHAI with slightly better fairness. BiasAware-ComHAI with severe bias: accuracy gap reduced from 24.2% (ComHAI) to 11.7% — a 12.5 pp improvement. FairPLACO under moderate bias: EO gap reduced from 26.1% to 11.0%.

Q: What is the limitation of your algorithms?

A: Three main limits: (1) The fairness correction is a post-hoc boost, not a principled Bayesian update — a better version would modify the prior itself. (2) Group-conditioned matrices need group labels at training time. (3) Lambda requires cross-validation — adds computational cost. Extending to multi-class K>2 and multiple protected attributes is future work.

TASK 3 ONE-LINE SUMMARY:

"We designed 5 novel fairness-aware Human-AI team algorithms. FairComHAI uses group-aware subset selection and fairness post-correction. BiasAware-ComHAI extends the confusion matrix to be group-conditioned, directly modelling and correcting human bias. Together, they reduce the accuracy gap from 24.2% to 11.7% under severe bias while maintaining near-ComHAI accuracy."